Sinoda Gebrehiwot Tesfay

Software Engineer

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- LinkedIn
- Github
- Portfolio

Summary

Software Engineer with 6 years of experience across backend, ML infrastructure, and full-stack development. Core strengths in Python, Rust, and Go, async API design, distributed systems, and real-time ML serving. Seeking a Backend or Platform Engineering role.

Skills

Languages
Python, Rust, Go, Java, TypeScript, JavaScript, Dart, SQL, RISC-V Assembly, FlutterFlow

Backend
FastAPI, Node.js, SQLAlchemy, REST, WebSockets, asyncio, Celery, Redis, SQLite, MongoDB ML & Systems: TensorFlow/Keras, scikit-learn, NumPy, distributed systems, Raft, MVCC, virtual memory, concurrency Frontend: React, Next.js, Flutter, Tailwind CSS, Unity DevOps & Testing: Docker, Kubernetes, GitHub Actions, CI/CD, pytest, JUnit 5, Git

Professional Experience

Esklate
Backend Engineer at Afro-Chat
2023 – 2026

- Directed the migration from Flask and MongoDB to an async FastAPI and SQLAlchemy architecture, reducing API latency by 35%.
- Built real-time ML serving infrastructure streaming live training telemetry to browsers at 20 fps over FastAPI WebSockets, with a SQLite experiment registry and on-demand model endpoints.
- Shipped the recommendation engine for Bazar, an AI-driven marketplace, using TensorFlow and scikit-learn fed by an automated Python and Selenium collection pipeline.

Kuraz Tech Engineering PLC
React Frontend Developer
2021 – 2023

- Cut application load times by 40% through code optimization and lazy loading, working within a 30-person frontend team.
- Drove the migration to Next.js with SSR and SSG, improving SEO and lifting user engagement by 25%.
- Mentored junior developers through code review and pair programming, reducing onboarding time by 30%

Esklate
Full Stack Developer (Internal Tooling)
2020 – 2021

- Built a Slack-integrated automation platform for Google Sheets workflows, adopted by 200+ employees across four time zones.
- Designed a timezone-aware notification system on Redis and Celery job queues, reducing missed deadlines by 30%.
- Implemented Kubernetes orchestration and highly available Go microservices for container monitoring, cutting deployment times by 60% at 99.9% uptime.

Education

Software Engineering
American College of Technology
07/2014 – 07/2019 | Addis Ababa, Ethiopia

Coding Academy (Data Structures & Algorithms)
A2SV
10/2022 – 10/2023 | Addis Ababa, Ethiopia

Projects

AI-RAD

Healthcare Screening & Records Platform

- Developed a secure web application that enabled hospitals to perform COVID-19 screenings, digitize patient records, and manage data through a centralized system, improving accessibility and operational efficiency.

Hospital COVID-19 screening and digital records platform with role-based access for doctors, nurses, and administrators. Won 1st place in the 2020 UNDP and MINT COVID-19 Innovation Challenge, securing a 580,000 ETB grant.

⦿ Tech stacks: React, JavaScript, Node.js, Firebase Authentication, Firestore, REST APIs, Role-Based Access Control, Responsive Web Design

RustV OS Lab

Bare-Metal RISC-V Operating System

- Implemented trap handling, Sv39 virtual memory, page tables, physical memory protection, system calls, and user-mode fault isolation using full register context frames. • Built a preemptive round-robin scheduler driven by CLINT timer interrupts, with assembly context switching and support for concurrent user-mode tasks.

Bare-metal RISC-V kernel in Rust (no_std) booting on QEMU without BIOS or SBI. Trap handling, Sv39 virtual memory, preemptive scheduling, user-mode syscalls, a from-scratch virtio-MMIO DMA block driver, and a persistent filesystem.

⦿ Tech stacks: Rust, RISC-V Assembly, QEMU, Sv39 Virtual Memory, Virtio-MMIO, DMA, TinyFS, UART, Cargo, GNU Make, GitHub Actions

GAN Observatory

Real-Time GAN Training Observatory

- Engineered a real-time GAN observatory that streams training telemetry to an interactive browser dashboard at approximately 20 FPS using FastAPI, WebSockets, and a background training engine.
- Implemented a dependency-light NumPy GAN from scratch with manual forward propagation, backpropagation, and Adam optimization, supporting live hyperparameter updates without restarting training.
- Built interactive visualizations for generated distributions, discriminator decision boundaries, learned feature spaces, RBF-MMD, precision, recall, and automatic mode-collapse detection.

- Developed reproducible experiment tracking and model serving with SQLite, allowing users to save configurations, seeds, metric histories, and generator weights, then serve new samples through REST APIs.
- Containerized the platform with Docker and established automated quality gates using GitHub Actions, pytest, and Ruff, while retaining TensorFlow/Keras support for larger MLP and DCGAN experiments.

⦿ Tech stacks: Python, NumPy, TensorFlow/Keras, FastAPI, WebSockets, SQLite, JavaScript, Docker, GitHub Actions

ImmuStore DB
Distributed Database Engine Built from Scratch
- Engineered a distributed database engine from scratch in pure Python, featuring a copy-on-write B+ tree storage core with O(log n) reads, writes, sorted scans, lazy record loading, and append-only persistence with no third-party runtime dependencies.

Distributed database engine written from scratch in pure Python: copy-on-write B+ tree storage, MVCC with shadow-paging commits and crash recovery, a Redis RESP async server, and Raft-based replication with leader election and failover.

⦿ Tech stacks: Python, asyncio, B+ Trees, MVCC, Raft Consensus, Redis RESP, TCP Networking, SQLite, Prometheus, Docker, Docker Compose, pytest, Ruff, GitHub Actions

More at github.com/yabtesfu